ORF 524 – Final Exam

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Contents

Notes:

- Duration of examination: *7 hours*.
- All parts are equally weighted (5 points).
- Please start each question on a separate page.
- Please provide as much detail as possible in your answers.
- Answers without proper justification will not receive (partial) credit.

1 Question 1: Probability Theory and Asymptotics (15 points)

1. Let $\{X_i : 1 \leq i \leq n\}$ be i.i.d. random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

   Let $M > 0$ be a finite constant. Show that

$$\mathbb{P}[|X_i| \leq M] = 1 \qquad \Longleftrightarrow \qquad \mathbb{P}\left[\frac{1}{n}\sum_{i=1}^{n}|X_i| \leq M\right] = 1.$$

   Note: The i.i.d. assumption is crucial and the result is not true in general.

2. Let $\{X_i : 1 \leq i \leq n\}$ be a random sample from $X \sim \mathbb{P}_\theta = \mathsf{Uniform}[0, \theta]$.

   Recall that $\bar{\theta}_n = 2\bar{X}_n$ is an unbiased estimator of θ. Show that

$$\hat{\theta}_n = \mathbb{E}[\bar{\theta}_n | X_{(n)}] = \frac{n+1}{n} X_{(n)}.$$

   Discuss this result in connection to the Rao-Blackwell Theorem and UMVU estimation.

   Hint: $\mathbb{E}[X_i | X_{(n)}, X_{(n)} \neq X_i] = X_{(n)}/2$ and $\mathbb{P}[X_{(n)} = X_i | X_{(n)}] = 1/n$.

3. Let $f(x) : \mathbb{R} \mapsto \mathbb{R}$ be a known function, and $a, b \in \mathbb{R}$ with $a < b$. Numerical integration is commonly used to approximate integrals of the form $I := \int_a^b f(x)dx$ when they are difficult (or impossible) to compute analytically. The basic procedure is as follows:

   Step 1: Construct S i.i.d. uniform random variables on $[a, b]$, denoted by $\{U_i : 1 \leq i \leq S\}$.

   Step 2: Approximate I by $I_S := \frac{1}{S}\sum_{i=1}^{S} f(U_i)(b - a)$.

   Give sufficient conditions so that $I_S \to_{\mathbb{P}} I$ as $S \to \infty$.

   Give an estimate of the uncertainty in the approximation.

2 Question 2: Optimal Point Estimation and Inference (50 points)

Suppose that $\{X_i : 1 \leq i \leq n\}$ is a random sample from $X \sim \mathbb{P}_\theta$, with $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$ and Lebesgue density

$$f_X(x;\theta) = a(\theta)\exp\left(-\frac{1}{2}x^2\right)\exp\big(b(\theta)x\big),$$

where $\theta \in \Theta = [0, \infty)$ is the unknown parameter, $a(\cdot)$ and $b(\cdot)$ are some functions.

1. Show that $a(\theta) = \frac{1}{\sqrt{2\pi}}\exp(-\theta/2)$ and $b(\theta) = \sqrt{\theta}$.

2. Consider $\bar{X}_n^2$ as an estimator of θ, where $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Give conditions on $\{X_i : 1 \leq i \leq n\}$ under which a method of moments estimator of θ exists and equals $\hat{\theta}_{\texttt{MM}} = \bar{X}_n^2$.
   Is $\hat{\theta}_{\texttt{MM}}$ unbiased?

3. Find the log likelihood function. Does θ admit a scalar sufficient statistic?

4. Show that the maximum likelihood estimator of θ is $\hat{\theta}_{\texttt{ML}} = \big(\max\{\bar{X}_n, 0\}\big)^2$.

5. Find the uniform minimum variance unbiased estimator $\hat{\theta}_{\texttt{UMVU}}$ of θ.

6. Show that the Cramér-Rao bound for unbiased estimators of θ is $\mathsf{CRLB}(\theta) = 4\theta/n$.
   Is $\mathsf{CRLB}(\theta)$ attained by the estimator $\hat{\theta}_{\texttt{UMVU}}$?
   Hint: If $Y \sim \mathsf{Normal}(\mu, \sigma^2)$, then $\mathbb{V}[Y^2] = 4\mu^2\sigma^2 + 2\sigma^4$.

7. Let $\theta_0 > 0$ be some constant and consider the one-sided testing problem:

$$\mathsf{H}_0 : \theta = \theta_0 \qquad \text{vs.} \qquad \mathsf{H}_1 : \theta > \theta_0.$$

   (a) Consider a test which rejects H_0 if and only if $\bar{X}_n > c$, where c is some constant (possibly depending on θ_0). Find c such that the testing procedure has 5% size.

   (b) Show that the testing procedure derived above is uniformly most powerful (within the class of tests of the same level).

   (c) Derive the power envelop of the 5% level hypothesis test above.

   (d) Construct a 95% confidence interval for θ based on the UMP test derived above.

3 Question 3: Misspecified Maximum Likelihood (35 points)

Let $\{X_i : 1, 2, \cdots, n\}$ be a random sample from $X \sim \mathbb{P}_{\theta_0}$ with $\mathbb{P}_{\theta_0} \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, where $F(x) = \mathbb{P}_\theta[X_i \leq x]$ is absolutely continuous with Lebesgue density $f(x;\theta)$. Let $f(x) = f(x;\theta_0)$ to save notation. Consider another family of Lebesgue densities, $\{g(x;\theta) : \theta \in \Theta \subseteq \mathbb{R}\}$, which may not be equal to $f(x;\theta)$. A *possibly misspecified* Maximum Likelihood Estimator (MLE) is given by

$$\breve{\theta}_n = \arg\max_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} \log g(X_i;\theta),$$

and the associated population optimization problem is

$$\breve{\theta}_0 = \arg\max_{\theta\in\Theta} \mathbb{E}_{\theta_0}[\log g(X;\theta)] = \arg\max_{\theta\in\Theta} \int [\log g(x;\theta)] f(x) dx.$$

assuming the "usual" regularity conditions hold. You will be asked to state and impose additional assumptions (or regularity conditions) as needed below.

1. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0}\left[\log\left(\frac{g(X;\theta)}{f(X)}\right)\right] = \int \log\left(\frac{g(x;\theta)}{f(x)}\right) f(x)dx \leq 0,$$

   with equality if only if $g(x;\theta) = f(x)$.

   Use this result to justify the following interpretation for the pseudo-true value $\breve{\theta}_0$:

$$\breve{\theta}_0 = \arg\min_{\theta\in\Theta} D(g(\cdot,\theta), f), \qquad D(g(\cdot,\theta), f) = \int [\log f(x)] f(x)dx - \int [\log g(x;\theta)] f(x)dx.$$

   The "metric" $D(g(\cdot,\theta), f)$ is called the *Kullbach-Leibler* (K-L) distance. Can you give an interpretation to the (possibly) misspecified MLE based on this criterion?

2. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0}\left[\frac{\partial}{\partial\theta} \log g(X;\theta)\Big|_{\theta=\breve{\theta}_0}\right] = 0.$$

   What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

3. Providing appropriate regularity conditions, show that

$$\sqrt{n}(\breve{\theta}_n - \breve{\theta}_0) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi(X_i) + o_{\mathbb{P}}(1) \to_d \mathcal{N}(0, V(\breve{\theta}_0)),$$

where $\mathbb{E}_{\theta_0}[\psi(X_i)] = 0$ and $\mathbb{E}_{\theta_0}[\psi(X_i)^2] = V(\breve{\theta}_0) = H(\breve{\theta}_0)^{-1}\Sigma(\breve{\theta}_0)H(\breve{\theta}_0)^{-1}$.

Give the exact form of the influence function $\psi(x)$.

Give the exact form of the variance components $H(\theta)$ and $\Sigma(\theta)$.

What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

4. Propose a consistent estimator of the asymptotic variance $V(\breve{\theta}_0)$.

Construct a feasible, asymptotically pivotal statistic.

What happens when $g(x;\breve{\theta}_0) = f(x)$ or when $g(x;\theta) = f(x;\theta)$?

5. Suppose that $X_i = \mathbb{1}(\theta_0 - \epsilon_i \geq 0)$ with $\{\epsilon_i : 1, 2, \cdots, n\}$ be a random sample from $\epsilon \sim \mathcal{N}(0,1)$.

   (a) Show that $\mathcal{P} = \{\mathsf{Bernoulli}(\Phi(\theta)) : \theta \in \Theta \subseteq \mathbb{R}\}$.

   Find the (correctly specified) maximum likelihood estimator $\hat{\theta}_{\mathtt{ML}}$ of θ_0.

   Find the asymptotic distribution of $\sqrt{n}(\hat{\theta}_{\mathtt{ML}} - \theta_0)$, and give the exact form of the asymptotic variance component(s).

   (b) Suppose a researcher employs the misspecified model $\epsilon \sim \mathsf{Uniform}(0,1)$.

   Find the misspecified maximum likelihood estimator $\breve{\theta}_n$ of $\breve{\theta}_0$, and compute the form of pseudo-true value $\breve{\theta}_0$ in this case.

   Find the asymptotic distribution of $\sqrt{n}(\breve{\theta}_n - \breve{\theta}_0)$, and give the exact form of the asymptotic variance component(s).